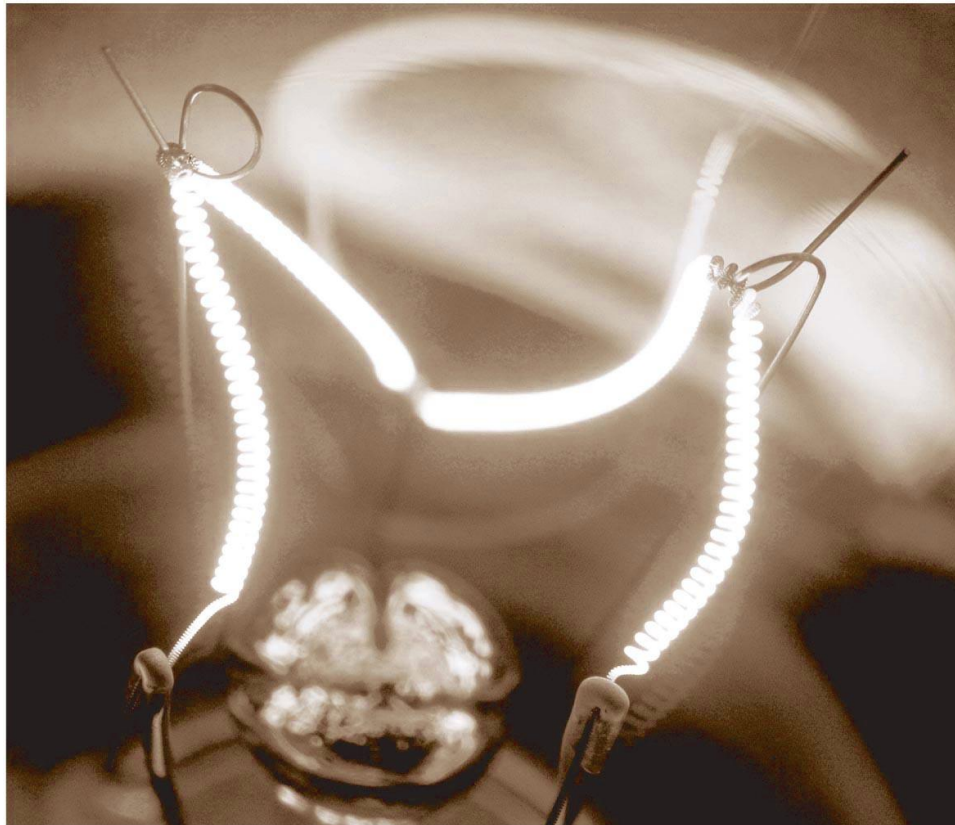


Ch26 Current and Resistance



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How does electric current heat this light bulb filament? Where does the energy come from?

Answer: electrons are accelerated by an electric field and collide with ions, so their kinetic energy is transferred into thermal energy of ions, and then into light energy.

André-Marie Ampère



André-Marie Ampère (1775-1836) was a French physicist and mathematician who was one of the founders of the science of classical electromagnetism. His name is one of the 72 names inscribed on the Eiffel Tower



TODAY'S TOPICS (CHAPTER 26)

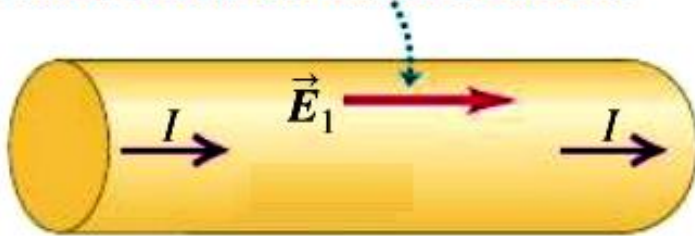
- ◆ Current and Current Density
- ◆ Resistance and Resistivity

E-field in a conductor?

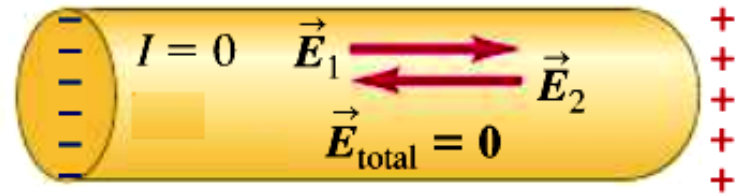
But we've learned that there is no electric field in a conductor...

At a equilibrium state

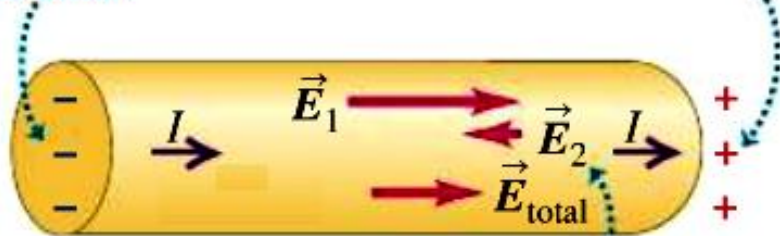
(a) An electric field $\vec{E}_1$ produced inside an isolated conductor causes a current.



(c) After a very short time $\vec{E}_2$ has the same magnitude as $\vec{E}_1$; then the total field is $\vec{E}_{\text{total}} = 0$ and the current stops completely.



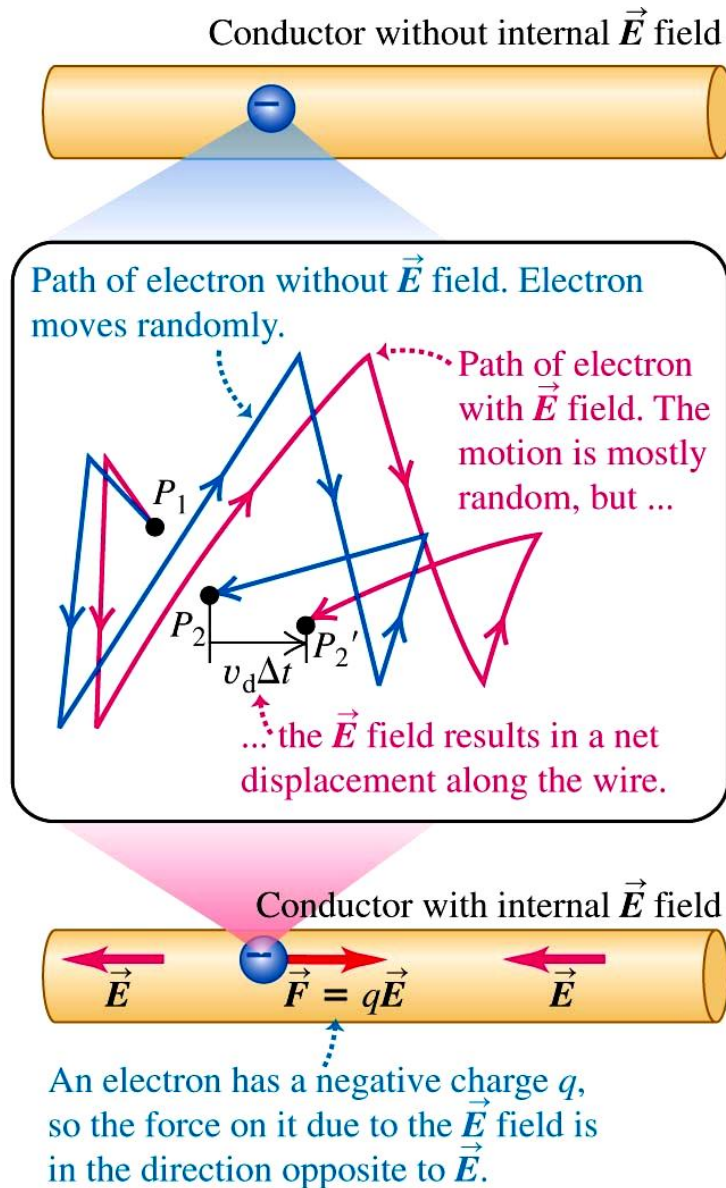
(b) The current causes charge to build up at the ends.



The charge buildup produces an opposing field $\vec{E}_2$, thus reducing the current.

Electric field is produced inside a conductor that is **not part of a complete circuit**, current flows for only **a very short time**.

What happens inside a conductor?

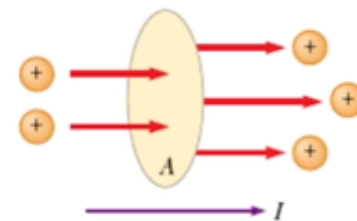


Random motion speed 10^6 m/s

Drift velocity $\vec{v}_d$ 10^{-4} m/s

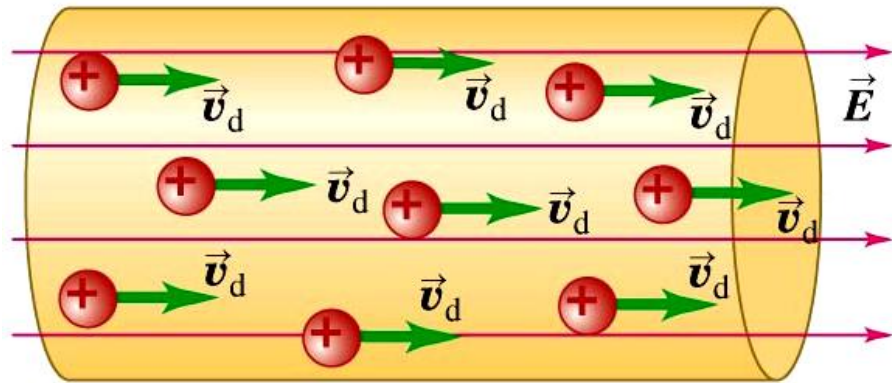
E-field is set up in the conductor at speed close to the speed of light

Current is the rate at which charge flows through some surface



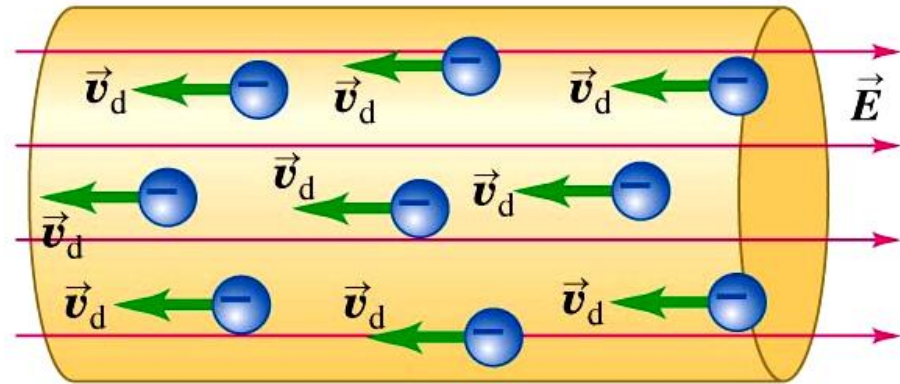
Conventional current

(a)



A **conventional current** is treated as a flow of positive charges, regardless of whether the free charges in the conductor are positive, negative, or both.

(b)



In a metallic conductor, the moving charges are electrons — but the *current* still points in the direction positive charges would flow.

$$i = \frac{dq}{dt} \quad (\text{definition of current}). \quad \text{Net charge per unit time}$$

The SI unit for current is the coulomb per second, or ampere (A):

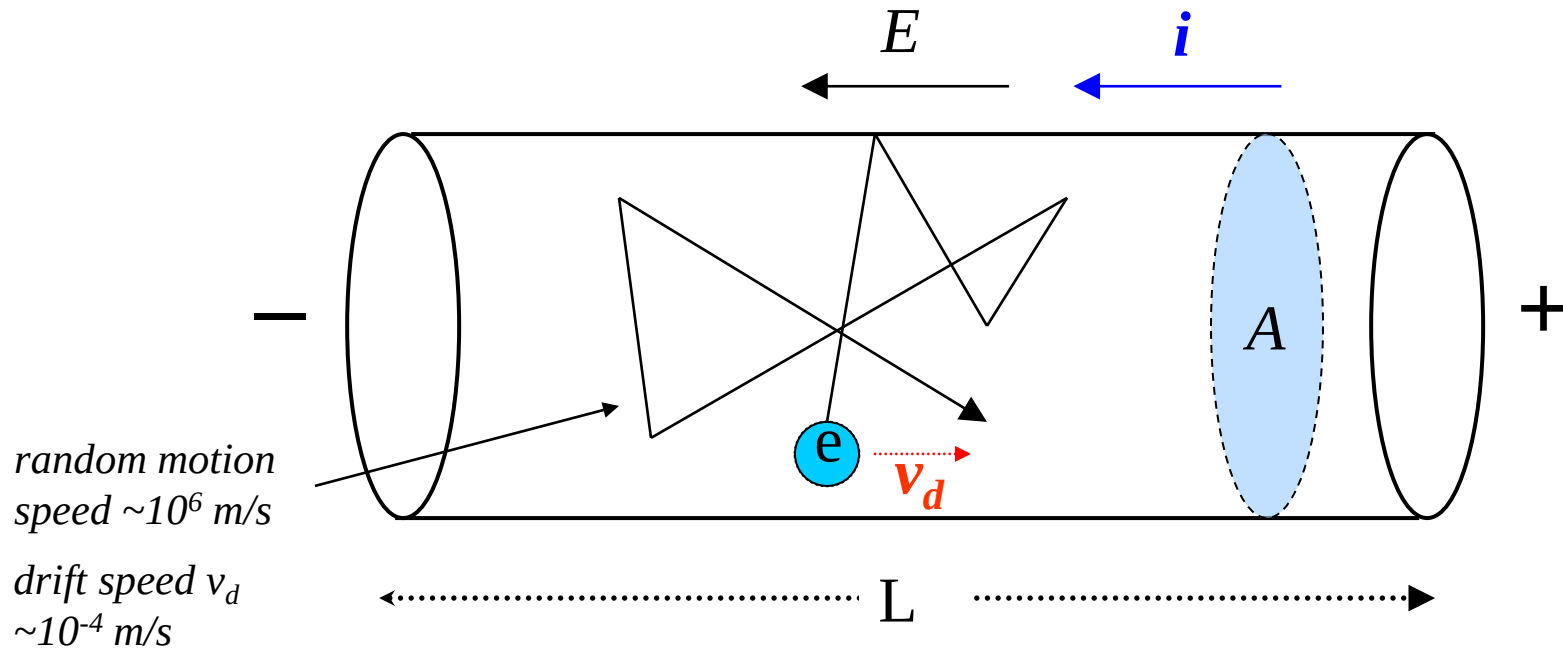
$$1 \text{ ampere} = 1 \text{ A} = 1 \text{ coulomb per second} = 1 \text{ C/s}.$$

SI UNIT

Previous (1946): 在真空中相距1米的两根横截面为圆形、粗度可忽略不计的无限长平行直导线，各通上相等的恒定电流，当两根导线之间每米长度所受力为 2×10^{-7} 牛顿时，各导线上的电流定义为1安培。

Now (2019): 由新的元电荷 $e = 1.602176634 \times 10^{-19} \text{ C}$ ($\text{C} = \text{A} \cdot \text{s}$) 和秒所定义

Current Density J and Drift speed v_d



$$q = (nAL)e \quad n = \text{number of } e^- \text{ per unit volume}$$

$$t = \frac{L}{v_d} \quad i = \frac{q}{t} = \frac{nALe}{L/v_d} = nAev_d$$

Current is a scalar quantity!

Current Density (电流密度)

Define **current density** $\vec{j}$ to be the current per unit cross-sectional area (unit : A/m²), that is

$$i = \int \vec{j} \cdot d\vec{A}$$

In most cases, the current is uniform across the surface and parallel to $d\vec{A}$, then $\vec{j}$ is also uniform and parallel to $d\vec{A}$

$$i = JA \quad \Rightarrow \quad J = \frac{i}{A}$$

$$i = nAev_d \quad \Rightarrow \quad \vec{j} = (ne)\vec{v_d}$$

$\vec{j}$ is a vector quantity! For negative carrier (electrons), J and v_d have opposite direction

Current Density:

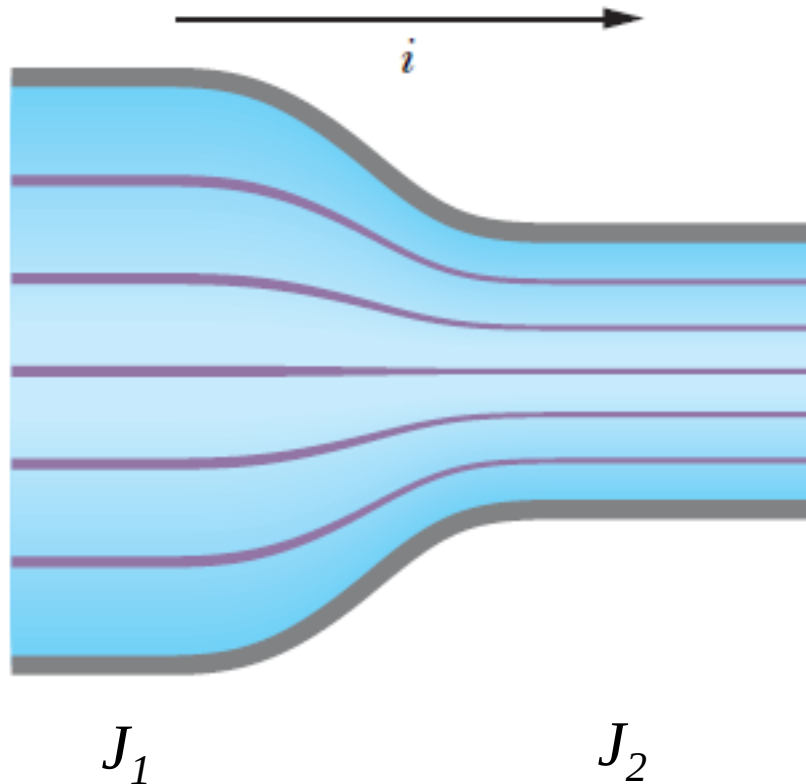


Figure shows how current density can be represented with a similar set of lines, which we can call *streamlines*.

The current, which is toward the right, makes a transition from the wider conductor at the left to the narrower conductor at the right. Since charge is conserved during the transition, the amount of charge and thus the amount of current cannot change.

However, the current density changes—it is greater in the narrower conductor.

current $\rightarrow$ no change in a circuit (charge conservation)

But current density $J_2 > J_1$

Resistance

Georg Ohm



Georg Simon Ohm (1789-1854), was a German physicist and mathematician. Using equipment of his own creation, Ohm found that there is a direct proportionality between the potential difference applied across a conductor and the resultant electric current. This relationship is known as Ohm's law (1827).



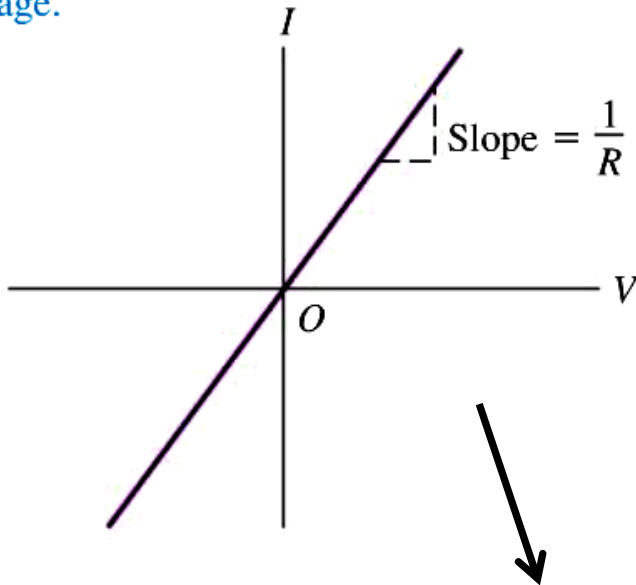
Henry Cavendish

Ohm's law was first discovered by Henry Cavendish, but Cavendish did not publish his electrical discoveries in his lifetime and they did not become known until 1879, long after Ohm had independently made the discovery and published himself. Thus the law came to bear the name of Ohm

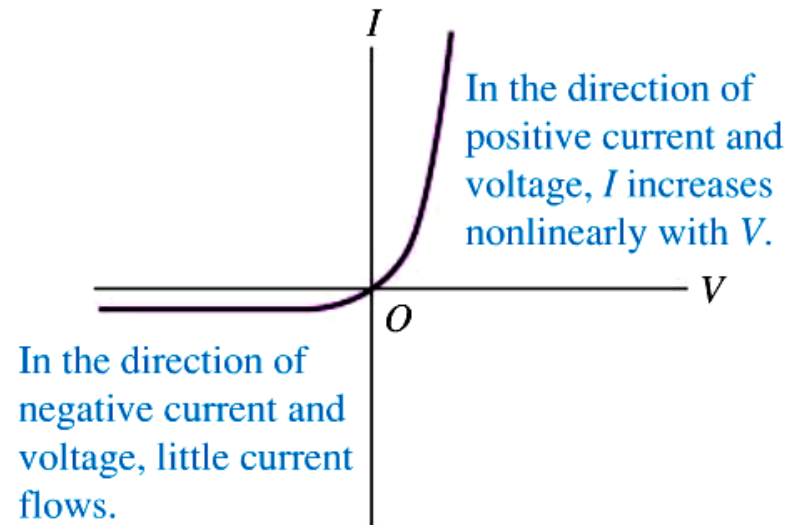
Ohm's Law

The statement $V = iR$ is not exactly Ohm's law !

Ohmic resistor (e.g., typical metal wire): At a given temperature, current is proportional to voltage.



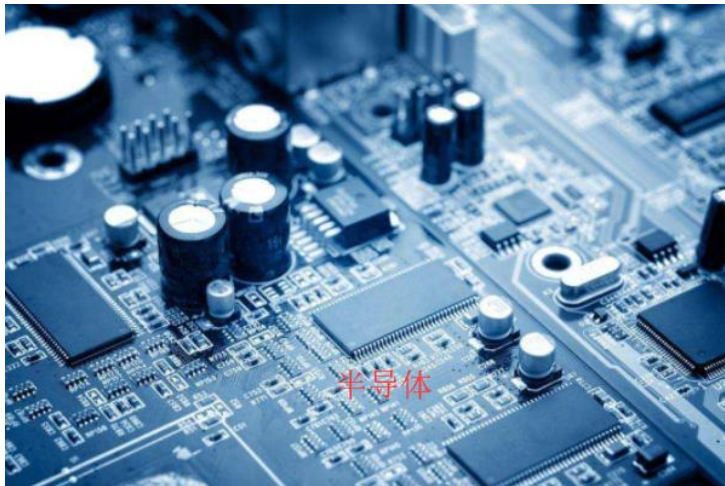
Semiconductor diode: a nonohmic resistor



- current through the device is always directly proportional to the potential difference applied to the device
- resistance of the device is independent of the magnitude and polarity of the applied potential difference

Resistivity – material dependent

- *Resistivity* of a material is the ratio of the electric field in the material to the current density it induces: $\rho = E/J$. The SI unit for ρ is $\Omega.m$.
- *The greater the resistivity, the greater the field needed*
- *Conductivity* is the reciprocal of the resistivity.



Resistivity 电阻率 and resistance 电阻

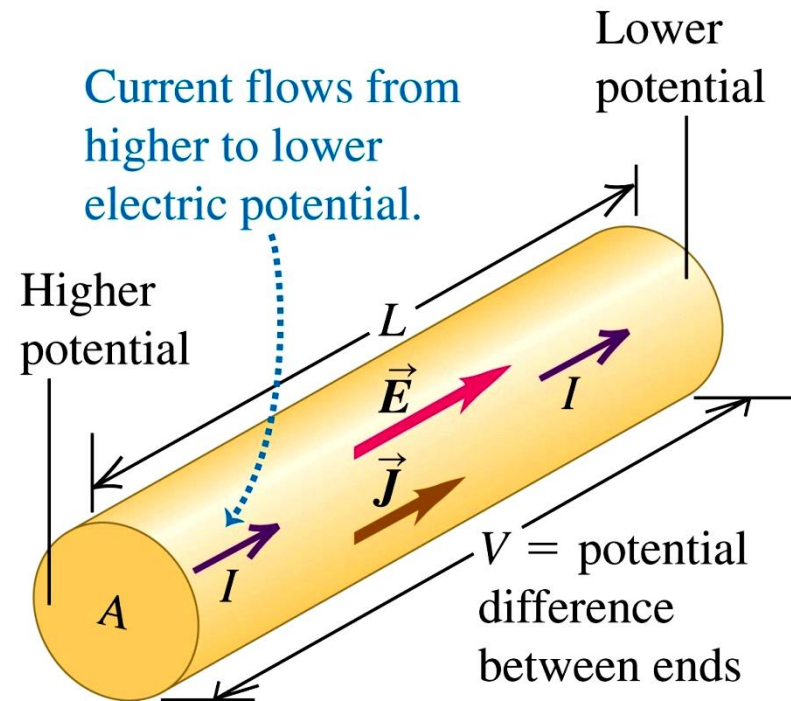
$$\vec{E} = \rho \vec{J}$$

$$V = EL \quad I = JA$$

$$\frac{V}{L} = \frac{\rho I}{A}$$

$$V = \frac{\rho L}{A} I$$

$$R = \frac{\rho L}{A} \quad (\text{relationship between resistance and resistivity})$$



Resistivity

Resistivities at Room Temperature (20 °C)

Substance		$\rho (\Omega \cdot \text{m})$
Conductors		
Metals	Silver	1.47×10^{-8}
	Copper	1.72×10^{-8}
	Gold	2.44×10^{-8}
	Aluminum	2.75×10^{-8}
	Tungsten	5.25×10^{-8}
	Steel	20×10^{-8}
	Lead	22×10^{-8}
	Mercury	95×10^{-8}
Alloys	Manganin (Cu 84%, Mn 12%, Ni 4%)	44×10^{-8}
	Constantan (Cu 60%, Ni 40%)	49×10^{-8}
	Nichrome	100×10^{-8}

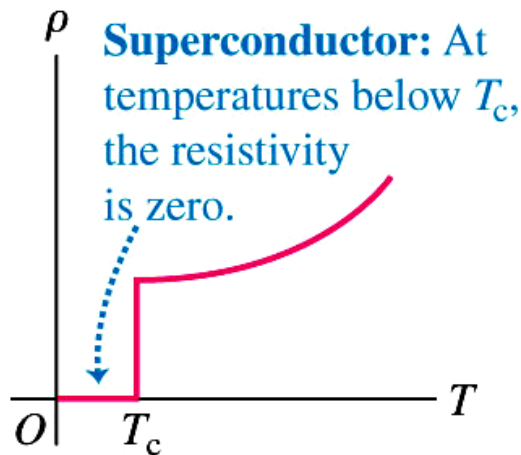
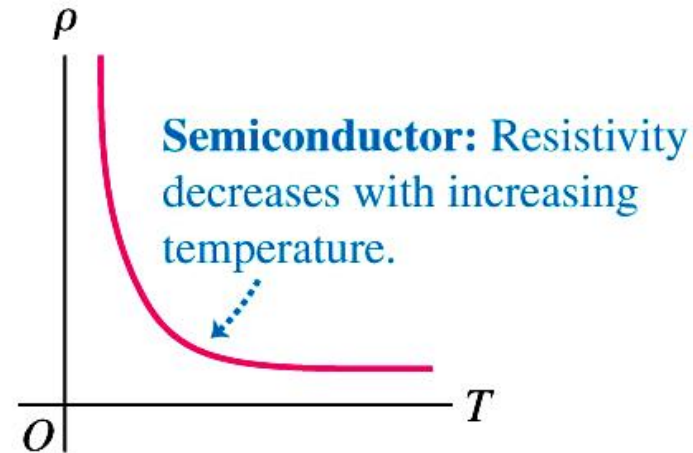
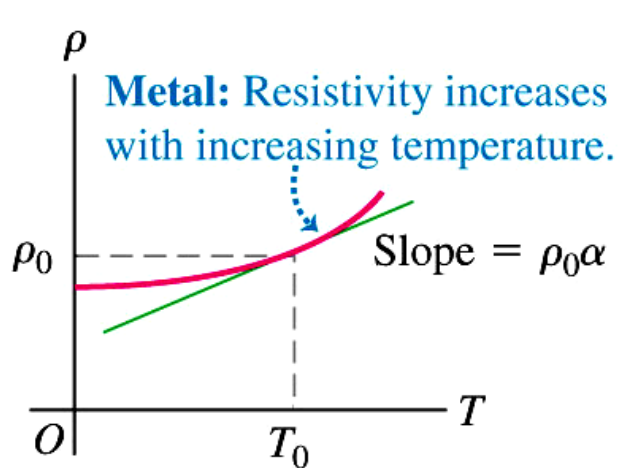
Resistivity

Substance	$\rho (\Omega \cdot \text{m})$
Semiconductors	
Pure carbon (graphite)	3.5×10^{-5}
Pure germanium	0.60
Pure silicon	2300
Insulators	
Amber	5×10^{14}
Glass	$10^{10} - 10^{14}$
Lucite	$> 10^{13}$
Mica	$10^{11} - 10^{15}$
Quartz (fused)	75×10^{16}
Sulfur	10^{15}
Teflon	$> 10^{13}$
Wood	$10^8 - 10^{11}$

Resistivity of different materials

Different materials' resistivity behaves qualitatively different when temperature changes.

For physicists, such qualitative changes are often more important than quantitative values.



1911 : Mercury at 4.2 K

Next 75 years, ~ 20 K

1986, ~ 40 K

1987, >77 K

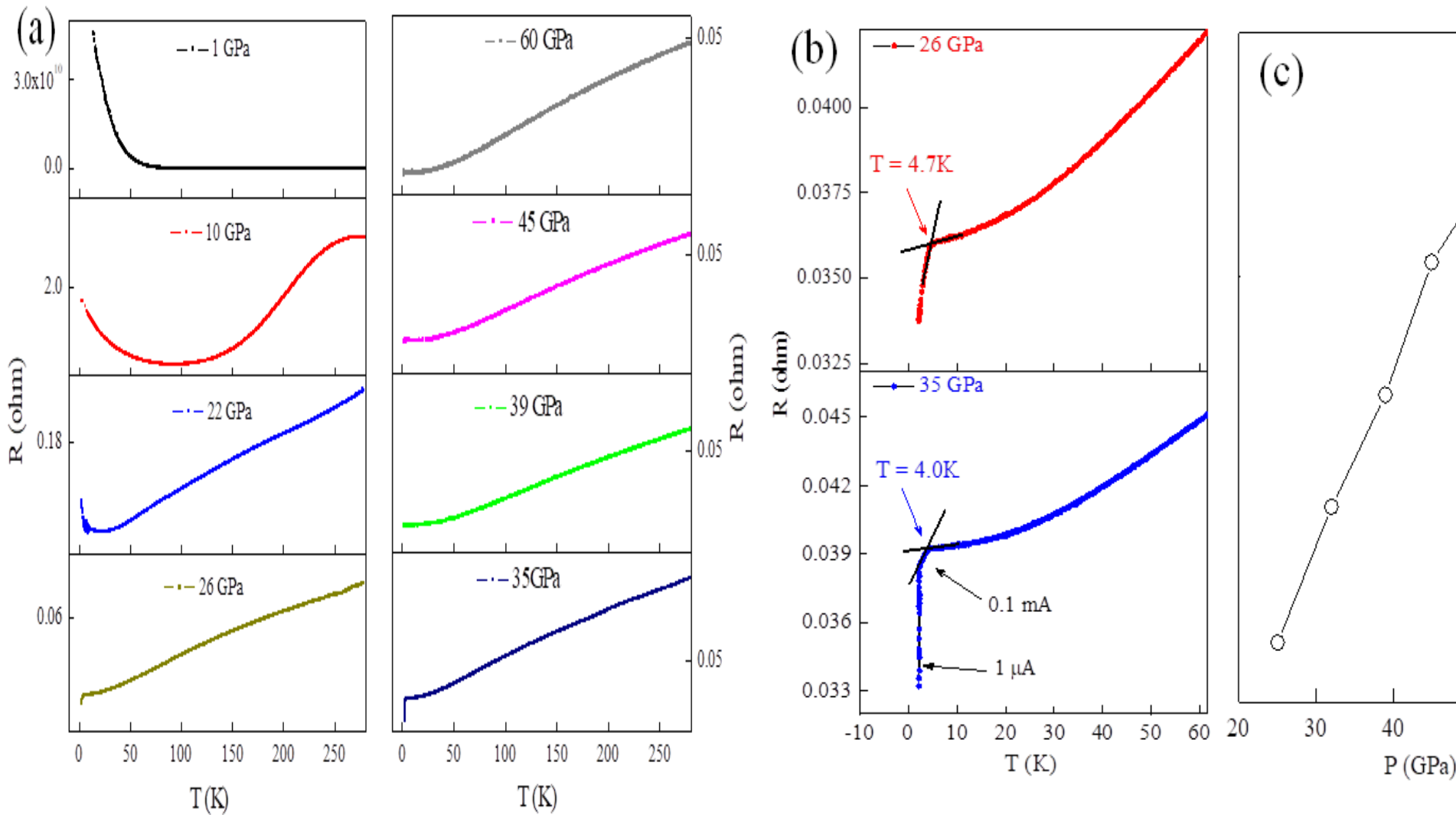
2010, 138 K

1994, 164 K HBCCO under high pressure

2015, 203 K H_2S and LaH_{10} under

2019, 283 K high pressure

Near RT superconductor!



My own work unpublished

Superconductivity at Room Temperature



Superconductivity in a hydride

Dasenbrock-Gammon^{2,6}, Raymond McBride^{1,6}, Mathew Debessai³,
An Vencatasamy², Keith V. Lawler⁴, Ashkan Salamat⁵ & Ranga P. Dias^{1,2}✉

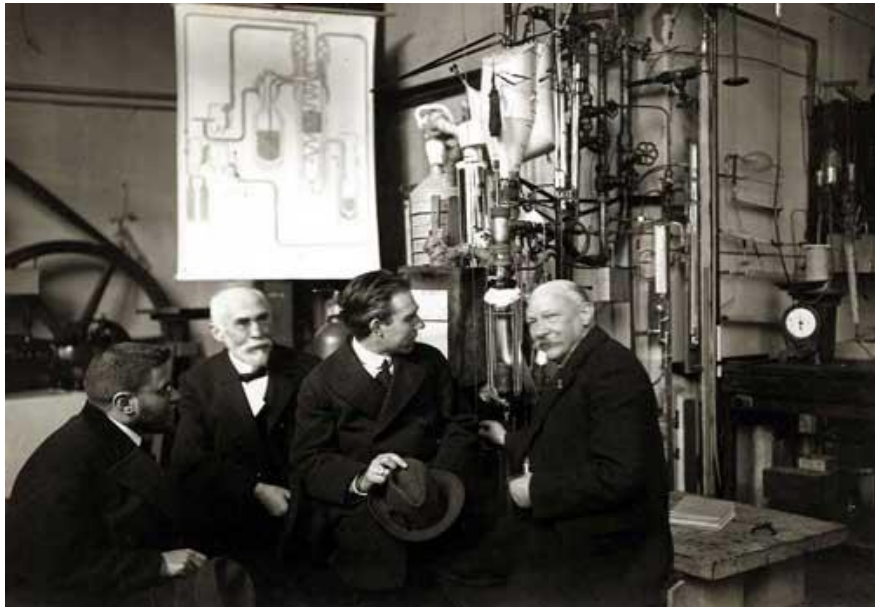
One of the longstanding challenges in experimental physics is the observation of superconductivity^{1,2}. Recently, high-temperature conventional superconductivity in hydrogen-rich materials has been reported in several systems

carbonaceous sulfur
hydride, C-S-H

$287.7 \pm 1.2 \text{ K}$

$@267 \pm 10 \text{ GPa}$

Superconductor



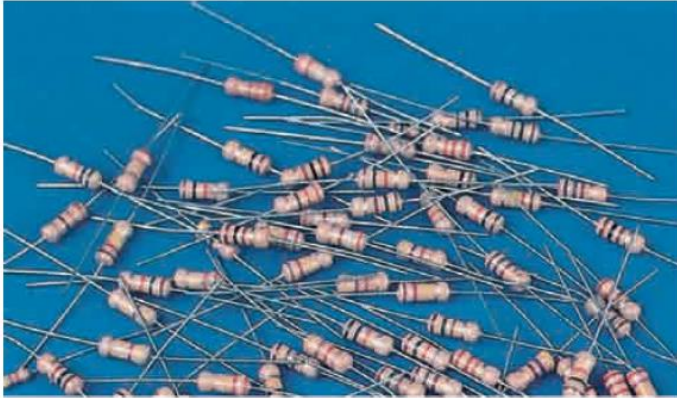
Superconductivity was discovered on April 8, 1911 by Heike Kamerlingh Onnes, who was studying the resistance of solid mercury at cryogenic temperatures using the recently produced liquid helium as a refrigerant.

1913: Nobel Prize

Explanation: 1957 BCS theory (1972 Nobel Prize). According to BCS, the superconductivity can only occur under 40 K. But... High-temperature superconductivity was observed!

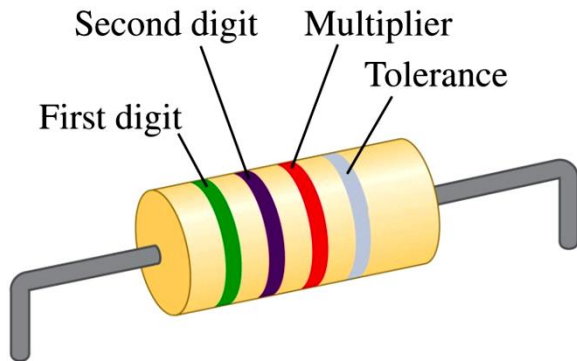
So here comes (one of) the most difficult problems: high-temperature superconductivity!

Resistor color codes



Color Codes for Resistors

Color	Value as Digit	Value as Multiplier
Black	0	1
Brown	1	10
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9



Tolerance : no band $\pm 20\%$, silver band $\pm 10\%$, gold band $\pm 5\%$

- This resistor has a resistance of $5.7 \text{ k}\Omega$ with a tolerance of $\pm 10\%$.

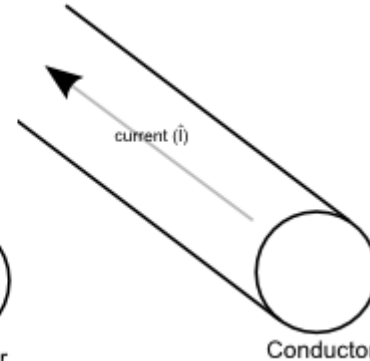
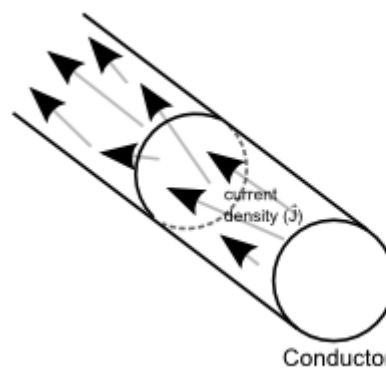
Resistance Calculator: <https://tool.520101.com/yuanqijian/5sehuandianzu/>

More on Resistivity and Resistance

Property of the material **VS** property of the material with a particular geometry

$$i = \int \vec{J} \cdot d\vec{A}$$

$$J = \frac{i}{A}$$



$\vec{J} \rightarrow$ current density

$i \rightarrow$ current

$$\rho = \frac{E}{J}$$

$$= \frac{V/L}{i/A}$$

$$R = \rho \frac{L}{A}$$

$$i \rho \frac{L}{A} = V$$

More on current density

$$J = \frac{i}{A}$$

$$\rho = \frac{E}{J}$$

$$i = \frac{q}{t} = \frac{nALe}{L/v_d} = nAev_d$$

$$J = (ne)v_d$$

$$v_d = a\tau$$

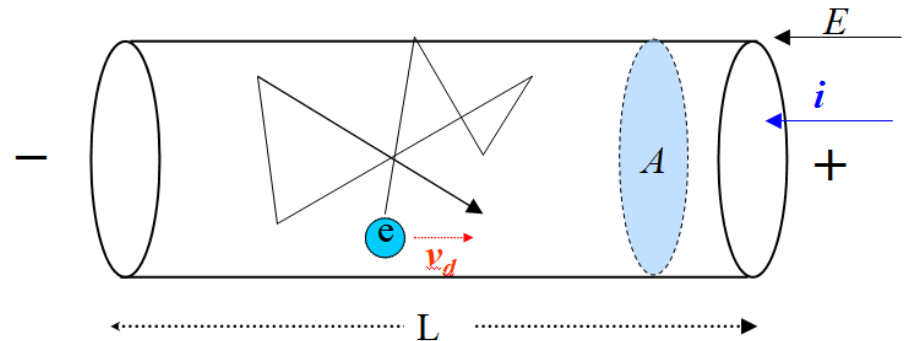
$$a = \frac{F}{m} = \frac{eE}{m}$$

τ average time between each collisions

$$J = (ne)v_d = (ne)a\tau = (ne)\frac{eE}{m}\tau$$

ρ is mostly dependent on random motion speed (as the drift speed is too small), we can say ρ is only related to intrinsic material properties.

$$\rho = \frac{m}{e^2 n \tau}$$



n = number of e per unit volume

random motion
speed ($\sim 10^6$ m/s)

Drift speed v_d
($\sim 10^{-4}$ m/s)

Quick check

The figure shows 3 pieces of wire.

(1) & (2) are made from the same material, while

(3) is made from material with twice the resistivity.

(1) & (3) have twice the diameter of (2), and have the same length



(1)



(2)



(3)

(a) Which has the highest resistance?

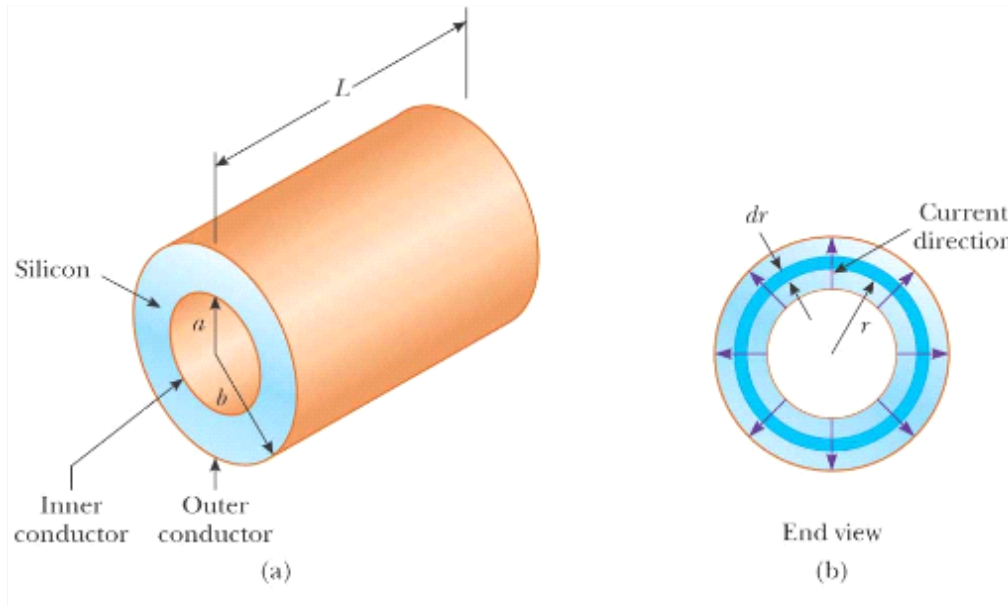
(2)

(b) If the same voltage is applied across each,
which will pass the highest current?

(1)

Resistance

Find resistance between inner and outer conductor.



$$dR = \rho \frac{dr}{A}$$

$$\text{But } A = 2\pi rL,$$

$$\therefore dR = \frac{\rho}{2\pi rL} dr$$

$$R = \int_a^b dR = \frac{\rho}{2\pi L} \int_a^b \frac{dr}{r} = \frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$$

Microscopic & Macroscopic Quantities in Ohm's Law

Microscopic	Macroscopic	Relation
$\mathbf{E}$	V	$V = \int \mathbf{E} \cdot d\mathbf{r} \quad \rightarrow \quad E L$
$\mathbf{J}$	I	$I = \int \mathbf{J} \cdot d\mathbf{A} \quad \rightarrow \quad J A$
ρ	R	$R = \frac{\rho L}{A}$
$\mathbf{J} = \frac{1}{\rho} \mathbf{E}$	$I = \frac{V}{R}$	

Power – The rate of energy being transferred

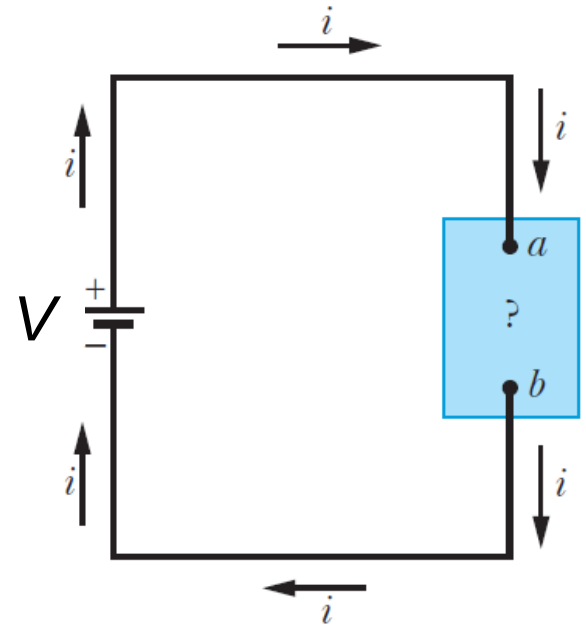
$$U = q V \quad (\text{energy})$$

$$dU = dq V \quad \boxed{i = \frac{dq}{dt}}$$
$$= i dt V$$

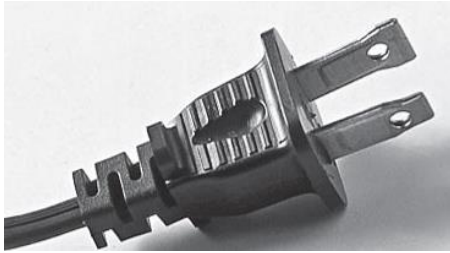
$$\frac{dU}{dt} = i V = P$$

SI unit : $V \cdot A$ or W , joule per sec, J/s

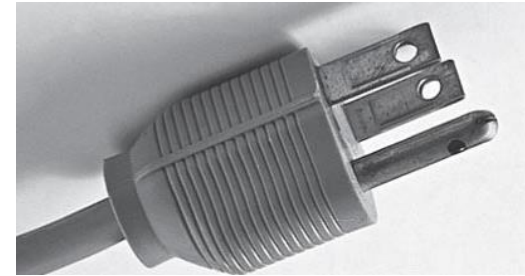
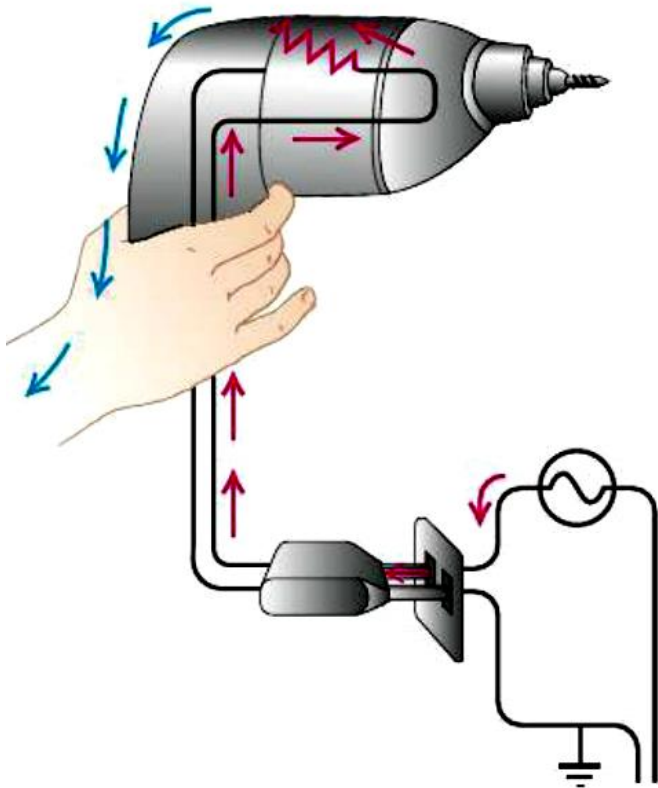
$P = i V$ applied to electrical energy transfers to all kinds of devices, while $P = i^2 R$ and $P = \frac{V^2}{R}$ only applied to the electric potential energy transfer to thermal energy in a device with resistance



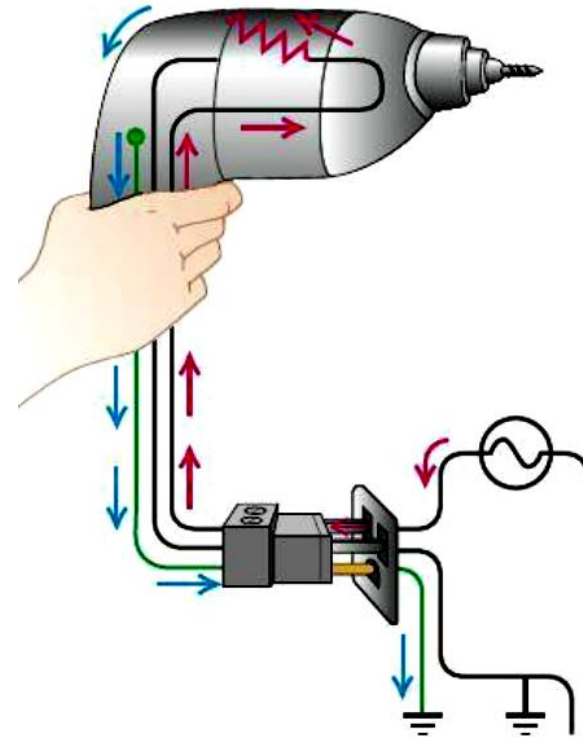
Electric Safety



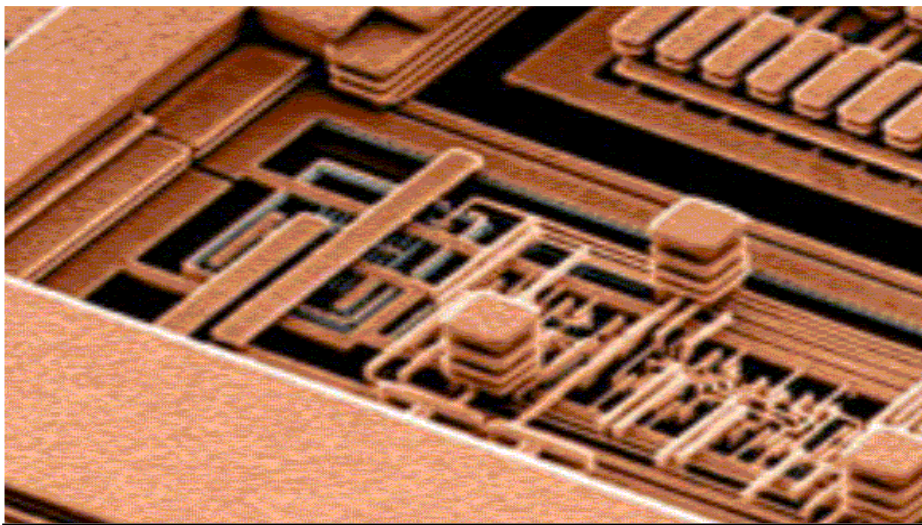
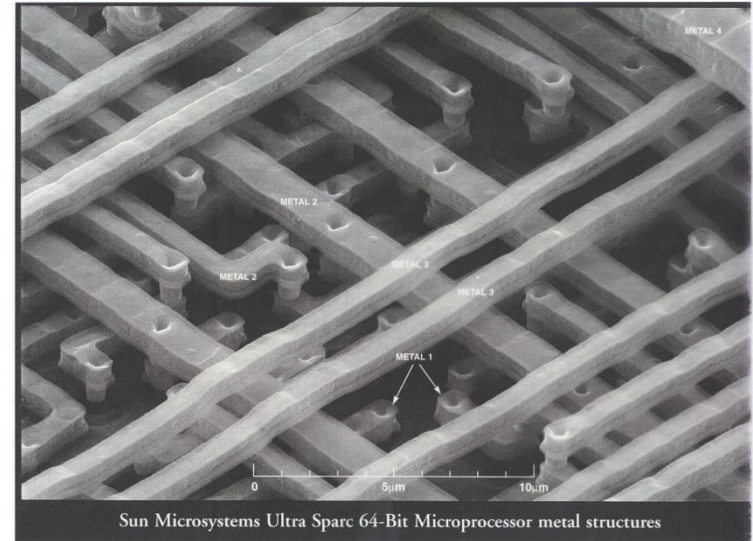
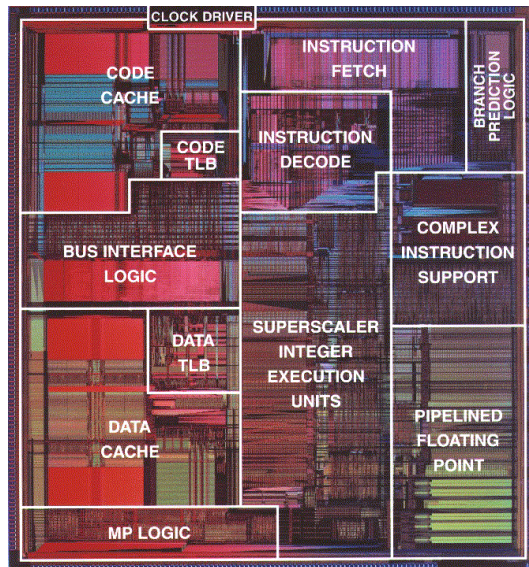
Two-prong plug



Three-prong plug



On-Chip Wirings on Si Semiconductor Devices



www.ibm.research.com

